Local-frame non-Kramers pyrochlore Hamiltonian and symmetry covariance

1 Local non-Kramers pseudospins

The rare-earth sites of the pyrochlore lattice are labelled by a Bravais cell R and a sublattice index $\mu = 0, 1, 2, 3$. Each sublattice carries its own local orthonormal frame

$$(\hat{\boldsymbol{x}}_\mu, \hat{\boldsymbol{y}}_\mu, \hat{\boldsymbol{z}}_\mu)\,, \qquad \hat{\boldsymbol{z}}_\mu \parallel \langle 111 \rangle, \tag{1}$$

with the four local Ising axes chosen as

$$\begin{aligned} \hat{\boldsymbol{z}}_0 &= \frac{(1,1,1)}{\sqrt{3}}, & \hat{\boldsymbol{z}}_1 &= \frac{(1,-1,-1)}{\sqrt{3}}, \\ \hat{\boldsymbol{z}}_2 &= \frac{(-1,1,-1)}{\sqrt{3}}, & \hat{\boldsymbol{z}}_3 &= \frac{(-1,-1,1)}{\sqrt{3}}. \end{aligned} \tag{2}$$

The low-energy doublet is represented by pseudospin-1/2 operators $S^\alpha_{R\mu}$ in this local frame. For a non-Kramers doublet, S^z transforms as the magnetic dipole component along the local Ising axis, whereas S^x and S^y transform as electric quadrupoles. Equivalently,

$$S^\pm_{R\mu} = S^x_{R\mu} \pm i S^y_{R\mu} \tag{3}$$

are local quadrupolar raising/lowering operators.

The key point for symmetry is that a space-group operation g does not act as a pure site permutation on the pseudospin components. It maps $(R, \mu) \mapsto (R', \mu')$ and rotates the local transverse frame by an angle $\chi_g(\mu)$ about the local $\hat{\boldsymbol{z}}$ axis. In the local basis one may write

$$\begin{aligned} g &: S^z_{R\mu} \mapsto S^z_{R'\mu'}, \\ g &: S^+_{R\mu} \mapsto e^{+i\chi_g(\mu)} S^+_{R'\mu'}, \\ g &: S^-_{R\mu} \mapsto e^{-i\chi_g(\mu)} S^-_{R'\mu'}. \end{aligned} \tag{4}$$

The sublattice-dependent phases are the local-frame expression of the standard non-Kramers pyrochlore symmetry action. A Hamiltonian written in local-frame operators is invariant when its bond and multi-spin phase factors compensate these local-frame rotations.

2 Nearest-neighbor non-Kramers XYZ Hamiltonian

The full nearest-neighbor non-Kramers pyrochlore Hamiltonian in the local pseudospin basis can be written as

$$H_{\mathrm{XYZ}} = \sum_{\langle ij \rangle} \Big[J_{zz} S^z_i S^z_j - J_\pm \left(S^+_i S^-_j + S^-_i S^+_j \right) + J_{\pm\pm} \left(\gamma_{ij} S^+_i S^+_j + \gamma^*_{ij} S^-_i S^-_j \right) \Big]. \tag{5}$$

Here $i \equiv (R, \mu)$ and $j \equiv (R', \nu)$, and γ_{ij} is the standard bond-dependent non-Kramers phase factor. In a common sublattice convention $\gamma_{ij} \in \{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$; the precise assignment depends on the oriented nearest-neighbor bond. This γ_{ij} factor is required because $S^{\pm}$ are local-frame quadrupolar operators. Under Eq. (4), $S_i^+ S_j^+$ acquires the phase $e^{i[\chi_g(\mu_i)+\chi_g(\mu_j)]}$, and the transformed bond coefficient $\gamma_{gi,gj}$ compensates this phase so that

$$\gamma_{ij} S_i^+ S_j^+ \mapsto \gamma_{gi,gj} S_{gi}^+ S_{gj}^+. \tag{6}$$

The Hermitian-conjugate $S^- S^-$ term transforms with the complex conjugate factor. Thus the complete sum in Eq. (5) is a scalar under the pyrochlore space group.

This form is often called the local-frame XYZ model. In Cartesian local coordinates it is equivalent to bond-dependent exchange in the transverse xy plane. The relation to local Cartesian couplings is, for the convention used by the present helper,

$$J_{\pm} = -\frac{J_{xx} + J_{yy}}{4}. \tag{7}$$

When $J_{xx} \neq J_{yy}$, the transverse anisotropy appears as a nonzero $J_{\pm\pm}$ channel with the sublattice/bond phase γ_{ij}.

2.1 XXZ limit used in the present scans

The numerical scans discussed here use the XXZ submanifold

$$J_{\pm\pm} = 0, \qquad H_{\mathrm{XXZ}} = \sum_{\langle ij \rangle} \left[J_{zz} S_i^z S_j^z - J_{\pm} \left(S_i^+ S_j^- + S_i^- S_j^+ \right) \right]. \tag{8}$$

For the $jpm03$ point, $J_{xx} = J_{yy} = 0.6 J_{zz}$, hence $J_{\pm} = -0.3 J_{zz}$ and $J_{\pm\pm} = 0$.

For ordinary periodic boundary conditions, all boundary twist phases are set to zero. More generally, the finite-cluster twist implementation multiplies the transverse hopping on a wrapped bond by the Peierls phase

$$S_i^+ S_j^- \mapsto e^{-i \boldsymbol{n}_{ij} \cdot \boldsymbol{\varphi}} S_i^+ S_j^-, \tag{9}$$

where $\boldsymbol{n}_{ij}$ is the integer wrap vector of the minimum-image bond. For the normal PBC calculations discussed here, $\boldsymbol{\varphi} = \mathbf{0}$.

In the local-frame convention, Eq. (8) is the $J_{\pm\pm} = 0$ scalar limit of Eq. (5). The full codebase contains non-Kramers $J_{\pm\pm}$ phase-factor logic in the general helper path, while `twist_trilinear_helper.py` deliberately writes the XXZ two-body limit plus the Kadowaki three-spin term used below.

3 Kadowaki three-spin term

The trilinear perturbation is the Kadowaki three-spin interaction

$$H_{3s} = \sum_{a=1}^{3} J_{3s,a} \sum_{\langle r, r', r'' \rangle_a} \left[e^{i\phi^{(a)}_{r,r'r''}} S_r^+ S_{r'}^z S_{r''}^z + e^{-i\phi^{(a)}_{r,r'r''}} S_r^- S_{r'}^z S_{r''}^z \right]. \tag{10}$$

The index $a = 1, 2, 3$ labels the three Kadowaki geometric classes of unordered nearest-neighbor pairs around the central site r:

- $a = 1$: collinear pair, where r' and r'' have the same sublattice index and lie on opposite tetrahedra sharing r;

- $a = 2$: same-tetrahedron pair, where r' and r'' are both neighbors of r in the same tetrahedron;
- $a = 3$: cross pair, where r' and r'' lie on opposite tetrahedra adjacent to r.

For each central site there are $3+6+6 = 15$ unordered pairs. The code supports either independent $(J_{3s,1}, J_{3s,2}, J_{3s,3})$ or a scalar J_3 broadcast to all three classes.

The phases are roots of unity,

$$e^{i\phi^{(a)}} = \omega^{p^{(a)}}, \qquad \omega = e^{2\pi i/3}, \tag{11}$$

with the integers $p^{(a)}$ read from the Kadowaki phase tables in the sublattice convention $\mu = 0, 1, 2, 3$. The implementation uses

$$\begin{array}{c|ccc} \mu_{\text{center}} & \text{outer sublattice } 1 & 2 & 3 \\ \hline 0 & -1 & +1 & 0 \end{array} \tag{12}$$

and the corresponding cyclic tables for the other central sublattices for the collinear class, together with the pair tables for the same-tetrahedron and cross classes. These phase tables are exactly what makes Eq. (10) a local-frame scalar. Under a space-group operation, the central transverse operator transforms as $S_r^+ \mapsto e^{i\chi_g} S_{gr}^+$, while the Kadowaki phase transforms as

$$e^{i\phi^{(a)}_{r,r'r''}} \mapsto e^{-i\chi_g(r)} e^{i\phi^{(a)}_{gr,gr',gr''}}, \tag{13}$$

so that the product $e^{i\phi^{(a)}} S^+ S^z S^z$ maps into the corresponding term in the same Hamiltonian. The Hermitian-conjugate S^- row transforms with the complex conjugate phase. Thus the complete sum over all triplets in Eq. (10) is invariant under the pyrochlore space group.

The present code writes the file order

$$S^z_{r'}\, S^{\pm}_r\, S^z_{r''}, \tag{14}$$

with the same complex coefficient as Eq. (10). Since the three operators act on distinct sites, this is the same operator product.

4 Quadrupolar strain probes

The thermal-expansion proxy is computed for five local-XY quadrupole operators:

$$Q_1, \quad Q_2, \quad Q_{xy}, \quad Q_{xz}, \quad Q_{yz}. \tag{15}$$

They are written as

$$Q = \sum_{R,\mu} \left[c^x_\mu(Q)\, S^x_{R\mu} + c^y_\mu(Q)\, S^y_{R\mu} \right], \tag{16}$$

with sublattice-dependent coefficients. The convention used in the operator files is

$$\begin{array}{c|cccc|cccc} & & c^x_\mu & & & & c^y_\mu & & \\ Q & \mu=0 & 1 & 2 & 3 & \mu=0 & 1 & 2 & 3 \\ \hline Q_1 & \sqrt{3} & \sqrt{3} & \sqrt{3} & \sqrt{3} & -1 & -1 & -1 & -1 \\ Q_2 & -1 & -1 & -1 & -1 & -\sqrt{3} & -\sqrt{3} & -\sqrt{3} & -\sqrt{3} \\ Q_{xy} & +\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & +\frac{1}{4} & +\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4} & +\frac{\sqrt{3}}{4} \\ Q_{xz} & +\frac{1}{4} & -\frac{1}{4} & +\frac{1}{4} & -\frac{1}{4} & -\frac{\sqrt{3}}{4} & +\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4} & +\frac{\sqrt{3}}{4} \\ Q_{yz} & -\frac{1}{2} & -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2} & 0 & 0 & 0 & 0 \end{array} \tag{17}$$

Thus, explicitly,

$$
\begin{aligned}
Q_1 &= \sum_{R,\mu} \left(\sqrt{3}\, S^x_{R\mu} - S^y_{R\mu} \right), \\
Q_2 &= -\sum_{R,\mu} \left(S^x_{R\mu} + \sqrt{3}\, S^y_{R\mu} \right),
\end{aligned} \tag{18}
$$

and

$$
\begin{aligned}
Q_{xy} &= \sum_R \left[\frac{1}{4} S^x_{R0} - \frac{1}{4} S^x_{R1} - \frac{1}{4} S^x_{R2} + \frac{1}{4} S^x_{R3} + \frac{\sqrt{3}}{4} S^y_{R0} - \frac{\sqrt{3}}{4} S^y_{R1} - \frac{\sqrt{3}}{4} S^y_{R2} + \frac{\sqrt{3}}{4} S^y_{R3} \right], \\
Q_{xz} &= \sum_R \left[\frac{1}{4} S^x_{R0} - \frac{1}{4} S^x_{R1} + \frac{1}{4} S^x_{R2} - \frac{1}{4} S^x_{R3} - \frac{\sqrt{3}}{4} S^y_{R0} + \frac{\sqrt{3}}{4} S^y_{R1} - \frac{\sqrt{3}}{4} S^y_{R2} + \frac{\sqrt{3}}{4} S^y_{R3} \right], \\
Q_{yz} &= \sum_R \left[-\frac{1}{2} S^x_{R0} - \frac{1}{2} S^x_{R1} + \frac{1}{2} S^x_{R2} + \frac{1}{2} S^x_{R3} \right].
\end{aligned} \tag{19}
$$

The signs above include the strain-operator convention used in `make_quad_operator_files.py`: $O_\mu = +Q_1$, $O_\nu = -Q_2$, and $O_{ij} = -\mathcal{Q}_{ij}$ relative to the manuscript quadrupoles.

For the legacy ED operator files, Eq. (**??**) is converted to raising/lowering rows through

$$
c^x S^x + c^y S^y = \left(\frac{c^x}{2} - \frac{ic^y}{2} \right) S^+ + \left(\frac{c^x}{2} + \frac{ic^y}{2} \right) S^-. \tag{20}
$$

The two E_g operators are uniform local transverse sums, while the three T_{2g} operators are sublattice-staggered local-XY combinations corresponding to Q_{xy}, Q_{xz}, and Q_{yz} shear channels. These operators transform as irreducible components of cubic strain. Consequently, a cubic-symmetric Hamiltonian cannot mix the E_g and T_{2g} response blocks.

The quantity plotted as a thermal-expansion coefficient is the connected energy–quadrupole covariance,

$$
\alpha_Q(T) = \frac{\langle QH \rangle_T - \langle Q \rangle_T \langle H \rangle_T}{T^2}. \tag{21}
$$

With the usual canonical derivative convention,

$$
\frac{d\langle Q \rangle_T}{dT} = \frac{\langle QH \rangle_T - \langle Q \rangle_T \langle H \rangle_T}{T^2}, \tag{22}
$$

so Eq. (17) is the temperature derivative of the thermal quadrupole expectation value in the same sign convention as the Hamiltonian.

5 Implementation status and caveat

For ordinary, non-symmetry-projected ED/FTLM runs, the generated Hamiltonian files used in the present scans implement the XXZ limit Eq. (8) and the Kadowaki term Eq. (10). In particular:

- the PBC two-body file `InterAll.dat` contains the $S^z S^z$ and $S^+ S^- +$ h.c. XXZ terms;
- the trilinear file `ThreeBodyG.dat` contains 480 rows for the 16-site PBC cluster when $J_3 \neq 0$, corresponding to 16×15 triplets and their Hermitian conjugates;
- the ordinary QED path prints `Loading three-body terms ... ThreeBodyG.dat`, confirming that the Kadowaki term is included in non-`--symm` calculations.

There is a separate computational caveat: the current QED streaming `--symm` path should not be used for the three-body Kadowaki Hamiltonian until `ThreeBodyG.dat` is both loaded and included in the automorphism validation. This caveat does not change the physical symmetry argument above; it only restricts which numerical execution path is safe.

Literature conventions

The local-frame non-Kramers pseudospin convention follows the standard pyrochlore treatments of Onoda–Tanaka, Savary–Balents, and related non-Kramers quantum spin ice literature, where S^z is dipolar and (S^x, S^y) are quadrupolar components. The three-spin interaction and its phase tables follow Kadowaki *et al.*, Phys. Rev. B **105**, 014439 (2022).